

PROJECT PROPOSAL

Data Structures & Algorithms with C++

Project and Team Information

Project Title

Personal Finance / Expense Splitter

Student / Team Information

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Proposal Description

Motivation

Managing shared expenses among friends, roommates, classmates, or travel groups is a common problem in everyday life. When multiple people contribute different amounts toward food, transportation, accommodation, shopping, or other group activities, it becomes difficult to accurately track who paid, how much each person owes, and whom they need to pay. Usually, people depend on manual calculations, notes, spreadsheets, or chat messages to keep track of these expenses. This often leads to confusion, calculation errors, forgotten payments, and unnecessary disagreements within the group.

Another major issue is the difficulty of settling balances efficiently. For example, if one person has paid ₹1,000, another ₹500, and others have paid smaller amounts, determining the minimum number of transactions required to settle everyone's balance can be time-consuming when done manually. Handling expenses in different currencies can make the calculation even more complicated, especially when members are travelling together.

The proposed Personal Finance/Expense Splitter aims to solve these problems by providing a simple system for recording users, groups, and transactions and automatically calculating individual balances. The system will determine how much each member owes or is owed and assist in simplifying the final settlement between group members.

The project will be implemented using Object-Oriented Programming concepts in C++, with separate classes for User, Group, and Transaction. Operator overloading will be used to make balance calculations and settlements more intuitive, while templates will provide flexible handling of different currencies and numerical values. The application is designed to be simple, relatable, and easy to demonstrate through real-life expense-sharing scenarios.

Project Goals and Milestones

The primary goal of the Personal Finance/Expense Splitter project is to develop a simple, user-friendly application that helps individuals and groups efficiently manage and settle shared expenses. The system aims to reduce manual calculations, minimize errors, and provide a clear understanding of each member's financial contribution and outstanding balance.

The major goals of the project are to:

- Develop an Object-Oriented C++ application using classes such as User, Group, and Transaction.

- Allow users to create groups and record expenses shared among multiple members.
- Automatically calculate each member's share, amount paid, amount owed, and net balance.
- Implement operator overloading to simplify and demonstrate balance calculation and settlement operations.
- Use templates for flexible handling of different currencies and numerical data types.
- Provide an efficient method for settling balances between group members with fewer transactions.
- Create a simple interface that makes the project easy to understand and demonstrate in real-life scenarios.

Project Milestones

#	Milestone
1	Requirement Analysis Identify common expense-sharing situations and finalize the features, inputs, and expected outputs.
2	System Design Design the class structure and relationships between User, Group, and Transaction. Plan the balance calculation and settlement logic.
3	Core Implementation Implement the basic C++ classes, constructors, data members, and functions for managing users, groups, and expenses.
4	Advanced OOP Features Implement operator overloading for balance operations and templates for currency handling.
5	Testing and Debugging Test the application with different group sizes, expenses, and currencies to identify and correct errors.
6	Final Demonstration

	Integrate all features, prepare sample real-life scenarios, document the project, and demonstrate the completed expense-splitting system.
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Project Approach

The Personal Finance/Expense Splitter will be implemented using the concept of C++ to demonstrate the difference between procedural and object-oriented approaches. The application will allow users to create groups, record shared expenses, calculate individual shares, maintain balances, and settle payments efficiently.

In the C++ implementation, the same functionality will be developed using Object-Oriented Programming. Separate classes such as “User”, “Group”, and “Transaction” will provide better organization and encapsulation. Vectors will be used for dynamic storage. Operator overloading will be implemented for balance calculations and comparisons, while templates will provide reusable support for different numeric types and currencies.

Algorithms Used

1. User/Group Management Algorithm: Add, search, display, and manage group members.
2. Expense Recording Algorithm: Store the payer, amount, participants, and currency for every transaction.
3. Expense Splitting Algorithm: Divide the expense among selected members and calculate each person's share.
4. Balance Calculation Algorithm: $Balance = Amount\ Paid - Amount\ Owed$
5. Debt Settlement Algorithm: Separate members into creditors and debtors and match them to determine who should pay whom.
6. Transaction Minimization Algorithm: Continue matching debtors with creditors until all balances become zero or sufficiently close to zero.
7. Currency Handling: Store currency information and use conversion logic where required. Templates in C++ will make the numerical implementation reusable.

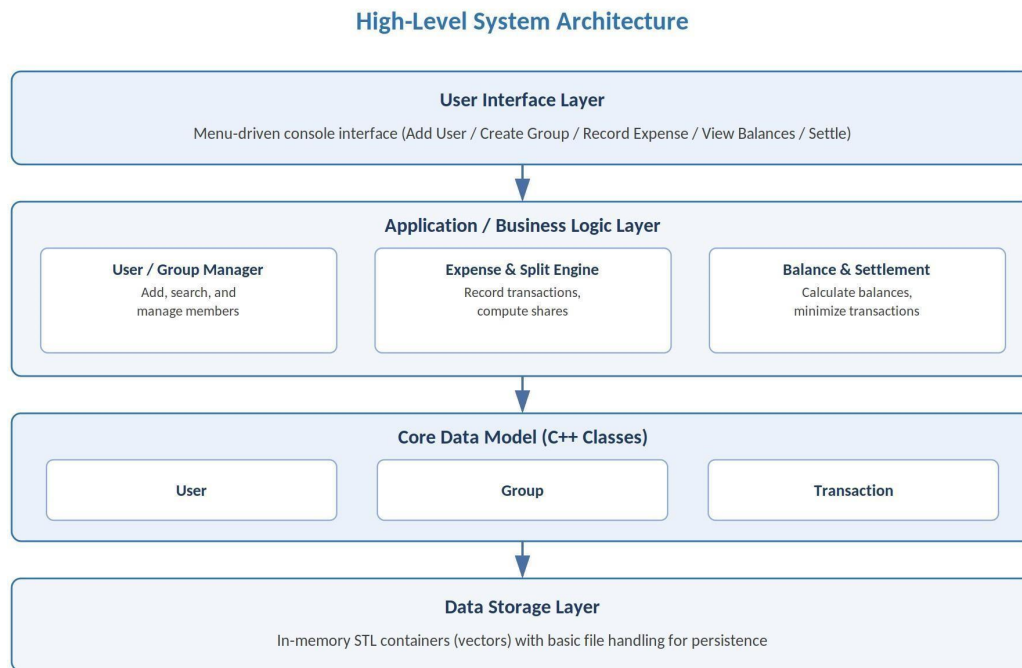
Technologies and Concepts

C++: Classes and objects, encapsulation, constructors, vectors, inheritance where applicable, operator overloading, templates, and STL.

Common: Expense-splitting logic, balance calculation, debt settlement, searching, sorting, and file-based data storage.

The project will be tested using relatable scenarios such as hostel expenses, restaurant bills, group trips, and shared shopping, making it simple to demonstrate the C++ implementations.

System Architecture (High Level Diagram)



Project Outcome / Deliverables

The Personal Finance/Expense Splitter will provide a simple and efficient solution for managing shared expenses among friends, roommates, classmates, and travel groups. The project will automatically calculate individual contributions, determine outstanding balances, and simplify the process of settling debts between members. This will reduce the need for manual calculations and minimize errors, confusion, and disputes related to shared payments.

The project will also demonstrate the practical application of Data Structures, Algorithms, C++ programming concepts. The C++ implementation will demonstrate Object-Oriented Programming, operator overloading, templates, and STL.

The system will be helpful in real-life situations such as hostel expenses, group trips, restaurant bills, rent sharing, and shopping. It will provide a clear record of who paid, who owes money, and the amount to be settled. Overall, the project will combine practical financial management with programming concepts to create a relatable, useful, and easy-to-demonstrate application.

Assumptions

The project assumes that all users involved in an expense are already registered in the respective group and that valid information is provided while entering transactions. Expenses are assumed to be divided equally among the selected members, with one person initially paying the complete amount. Currency conversion, if required, will use predefined exchange rates rather than live rates. The system will not consider interest, penalties, or additional charges. User IDs are assumed to be unique for accurate transaction tracking. Data may be stored during program execution or through basic file handling.

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DSA with C++

Team ID:DSCPP-III-2026-T272

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